Project Design Phase-I
Solution Architecture

Date	28 October 2026
Team ID	Individual Project
Project Name	**Health360_HealthCare_Analysis**
Maximum Marks	4 Marks

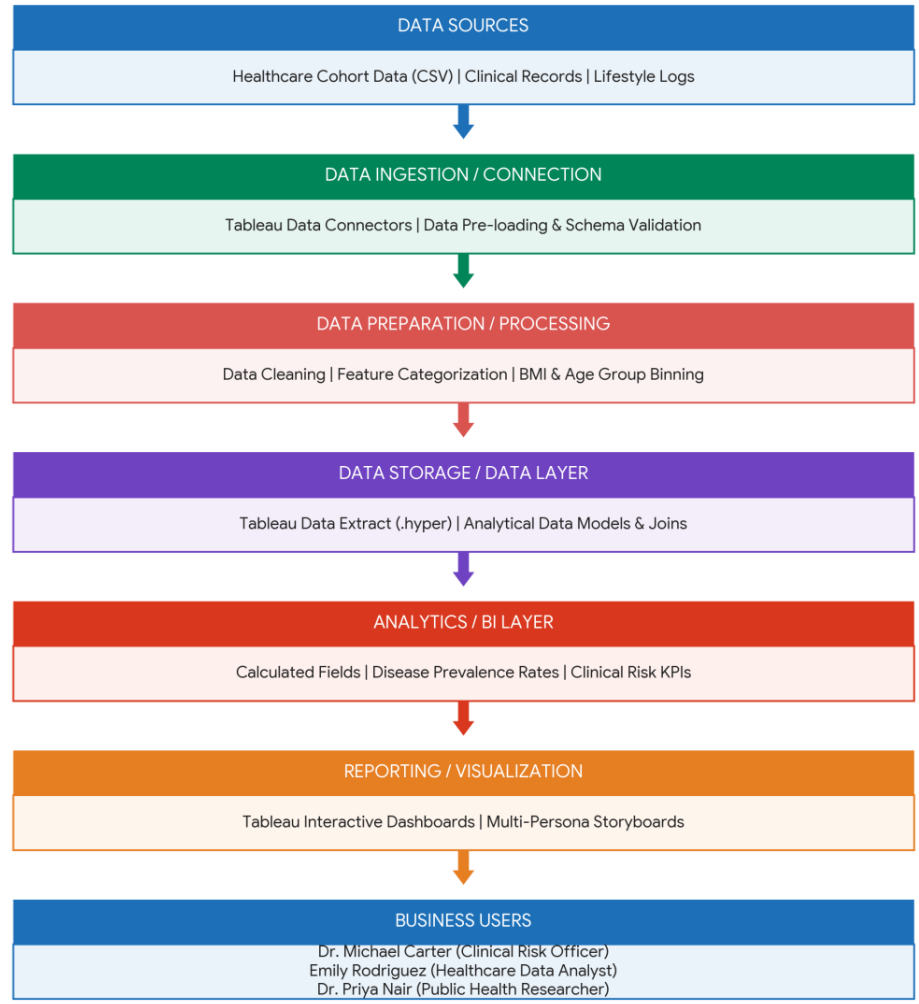
Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

SOLUTION ARCHITECTURE



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Sources	Stores raw clinical metrics (blood pressure, glucose, BMI, cholesterol) and lifestyle indicators (exercise level, smoking status) for N = 5,000 patient cohort.	CSV File / Healthcare Database
Data Ingestion / Connection	Establishes reliable connection to raw dataset, handles field mapping, data loading, and structural schema validation.	Tableau Data Connector / File Ingestion
Data Preparation / Processing	Cleans missing data, handles data formatting, standardizes attributes, and creates calculated demographic & risk bins (e.g., Obese, Overweight, Normal, Underweight).	Tableau Prep / Python (pandas)
Data Storage / Data Layer	Stores processed extracts optimized for fast visual querying, aggregation, and multi-variable filtering.	Tableau Data Extract (.hyper)
Analytics / BI Layer	Computes core KPI aggregations and disease prevalence metrics (Diabetes: 34.92%, Heart Disease: 29.98%, Stroke: 24.86%) and evaluates clinical threshold parameters.	Calculated Fields / Aggregations
Reporting / Visualization	Renders interactive Tableau dashboards, scatter plots, correlation heatmaps, and guided Tableau Stories tailored to user scenarios.	Tableau Desktop / Tableau Public
Business Users	Provides persona-centric visual insights for Dr. Michael Carter (Clinical Risk Assessment Officer), Emily Rodriguez (Data Analyst), and Dr. Priya Nair (Public Health Researcher).	Interactive Web / Cloud Dashboard